

MANJIMUP, Australia

WMO: 946170

Lat: 34.2505S

Lon: 116.1450E

Elev: 942

StdP: 14.20

Time Zone: 8.00 (E08)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.5	40.9	35.2	31.1	53.1	36.9	33.3	50.5	24.5	57.6	21.8	57.4	6.3	20	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	23.1	91.9	66.1	87.9	65.3	84.1	64.3	69.0	83.7	67.4	81.4	66.0	79.1	9.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.9	95.4	71.7	63.3	90.1	70.2	61.9	85.8	69.1	33.7	83.4	32.4	81.6	31.2	79.0	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	17.6	15.8	DB	36.2	100.4	1.0	2.9	35.4	102.5	34.8	104.3	34.2	105.9	33.5	108.0	(4)
(5)			WB	35.5	72.6	1.3	2.4	34.6	74.3	33.9	75.7	33.2	77.1	32.3	78.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	59.1	67.8	68.3	65.8	60.9	56.5	52.7	50.9	51.6	53.0	56.7	61.5	64.8	(6)
(7)		DBStd	7.96	5.69	5.84	6.17	5.32	4.19	3.37	3.25	3.73	4.31	5.05	5.85	6.17	(7)
(8)		HDD50	87	0	0	0	0	3	11	29	25	16	3	0	0	(8)
(9)		HDD65	2603	33	29	63	147	267	369	438	416	360	262	139	80	(9)
(10)		CDD50	3426	551	514	490	327	203	92	56	74	106	212	344	457	(10)
(11)		CDD65	465	119	123	88	23	2	0	0	0	0	6	32	72	(11)
(12)		CDH74	4744	1239	1160	818	171	13	0	0	0	10	87	407	839	(12)
(13)		CDH80	1814	511	465	307	36	0	0	0	0	0	14	139	342	(13)
(14)	Wind	WSAvg	8.8	9.4	9.3	8.6	7.7	7.8	8.3	8.6	8.9	9.4	8.9	9.0	9.1	(14)
(15)	Precipitation	PrecAvg	38.6	0.8	0.7	1.2	2.3	4.9	6.2	7.1	5.7	4.1	2.8	1.9	0.9	(15)
(16)		PrecMax	50.0	3.6	2.4	4.9	7.6	10.2	10.9	11.4	9.5	7.2	5.6	4.8	3.0	(16)
(17)		PrecMin	27.4	0.0	0.0	0.2	0.2	1.9	0.6	4.0	2.0	1.5	0.4	0.2	0.1	(17)
(18)		PrecStd	5.4	0.9	0.6	1.0	1.5	2.0	2.4	2.1	1.7	1.5	1.3	1.2	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.2	95.2	95.1	84.9	75.6	66.6	64.6	68.3	74.7	81.8	90.7	95.8	(19)
(20)			MCWB	66.2	67.9	65.4	62.8	58.5	55.0	55.1	55.2	57.4	62.5	63.4	65.3	(20)
(21)		2%	DB	90.3	90.1	87.8	78.8	70.6	63.7	61.6	63.8	68.4	75.8	84.2	89.0	(21)
(22)			MCWB	66.4	66.6	65.4	62.4	57.5	55.3	54.8	55.3	56.8	60.2	62.4	65.1	(22)
(23)		5%	DB	85.6	85.8	82.4	74.3	66.9	61.2	59.2	61.0	64.2	70.7	78.5	83.0	(23)
(24)			MCWB	65.5	65.3	64.4	60.7	57.6	54.8	54.0	54.5	55.5	58.6	61.3	63.4	(24)
(25)		10%	DB	81.3	81.6	77.9	70.3	63.9	59.2	57.3	58.5	61.0	66.5	73.4	77.7	(25)
(26)			MCWB	64.2	64.4	62.6	59.8	56.8	54.1	52.9	53.2	54.3	56.9	59.7	62.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.9	72.0	69.3	66.3	63.3	59.8	59.0	59.6	60.7	64.8	66.5	69.6	(27)
(28)			MCDWB	85.9	86.3	83.4	75.5	67.8	62.2	60.8	62.8	67.8	76.5	82.8	86.0	(28)
(29)		2%	WB	68.9	69.0	67.5	64.3	60.8	57.8	56.7	57.3	58.4	61.9	64.3	66.6	(29)
(30)			MCDWB	83.3	83.4	80.7	71.9	65.9	60.6	59.4	60.6	65.0	72.3	77.8	82.9	(30)
(31)		5%	WB	67.1	67.1	65.9	62.9	59.3	56.2	54.9	55.5	56.6	59.7	62.6	64.7	(31)
(32)			MCDWB	80.7	80.5	77.1	70.6	63.8	59.3	57.7	59.5	62.4	68.2	74.8	79.4	(32)
(33)		10%	WB	65.2	65.5	64.2	61.3	58.0	54.8	53.4	54.0	54.9	57.7	60.8	62.7	(33)
(34)			MCDWB	78.4	78.0	74.0	68.2	62.4	57.9	56.5	57.9	60.2	64.7	71.9	75.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	23.5	23.1	20.9	17.5	15.1	12.9	12.9	13.8	15.6	18.0	20.5	22.2	(35)
(36)			MCDBR	32.4	31.2	29.2	24.2	19.7	15.8	15.4	17.2	21.2	26.2	30.3	32.5	(36)
(37)			MCWBR	11.6	10.5	10.1	9.9	9.9	9.8	9.6	9.7	10.8	12.1	12.0	11.7	(37)
(38)		5% WB	MCDBR	27.3	26.3	22.8	19.5	15.3	13.3	13.1	14.7	18.4	23.0	25.8	28.1	(38)
(39)			MCWBR	11.2	10.1	9.3	9.1	9.3	9.6	9.5	9.6	10.5	11.4	11.2	11.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.323	0.322	0.314	0.305	0.297	0.297	0.293	0.301	0.322	0.317	0.321	0.318	(40)	
(41)		taud	2.599	2.612	2.628	2.652	2.647	2.622	2.632	2.598	2.509	2.547	2.563	2.596	(41)	
(42)		Ebn at Noon	322	316	307	293	279	269	277	290	299	313	320	325	(42)	
(43)		Edh at Noon	33	31	29	25	22	21	22	26	31	33	34	33	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2418	2132	1694	1208	884	719	776	1023	1371	1783	2158	2397	(44)	
(45)		RadStd	148	99	110	74	63	46	35	54	69	112	134	135	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-142	N/A	N/A	(47)

Nomenclature: See separate page